Evaluating Controls for Trading Compliance: An Offline Study of Order Splitting and Cancellation Timing

Scripted experiments on surveillance and pre-trade gates

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Abstract

An order-level size limit can admit a large displayed quantity when it is split across small orders. We study the evaluation of this failure and its repair in a scripted limit-order-book simulator. Six 10-lot orders pass the original gate and detectors in all 40 tested seeds. An aggregate cap removes their response-dependent gain at the original settings, but that result depends on the market-response threshold. In a separate cap-threshold grid with the cancellation throttle disabled, the same 20-lot cap yields a mean responsive-minus-unresponsive wealth difference of 0.00 tick-lots at threshold 0.4 and 48.00 at threshold 0.2. Throttle tests over 12-minute and one-hour sessions distinguish persistent lockout from recovery after old submissions leave the window. The full aggregate policy reduces maker volume by 14.1% and 28.2% in two short-session controls and by 75.2% in the one-hour test. Finally, a legitimate maker produces alerts in 34 of 40 episodes when its re-quote cadence meets the detector's 180-second window. These are configuration-dependent regression cases for compliance-control testing. The study uses synthetic prices and fixed trading scripts; it does not estimate real-market detection accuracy or evaluate LLM traders.

1. Introduction

A gate that rejects a 60-lot order can still admit six 10-lot orders at the same price. Each submission passes the size check, although the resulting display can influence the market in the same way. A cap on total same-side quantity addresses this implementation gap. Whether that change provides useful protection depends on what the market responds to and what legitimate trading the cap prevents.

We examine that question in a small, replayable trading simulator. The study separates three outcomes: whether a surveillance rule flags a sequence, whether a pre-trade gate removes its benefit from the simulated price response, and how much legitimate execution the gate obstructs. An alert and an order rejection measure different events. Neither, by itself, establishes economic prevention or legal compliance.

Three features of the evaluation turn out to matter. First, the aggregate cap can keep the book below the threshold that activates the stipulated price response. Zero gain in that setting is therefore weaker evidence than it initially appears. Second, a size cap changes the number of accepted submissions and hence the activation of a cancellation throttle. Third, a detector window excludes some legitimate quoting schedules by construction; moving the schedule inside that window can produce alerts without changing the trading objective.

The contribution is a set of controlled comparisons and regression cases for testing candidate trading-compliance controls. The scripts, labels, gates, and detectors were developed by the same author, with assistance disclosed in Section 10. The environment originated in a planned study of LLM trading agents, but the reported trading actions are entirely scripted. The present study evaluates the test environment and its controls, not model behavior.

2. Related work

Published research already establishes that order-book information can make traders vulnerable to spoofing. Wang and Wellman [1] study this interaction with trading agents; Wang et al. [7] extend that work with strategic responses and market-design changes. Cartea, Jaimungal, and Wang [18] model a spoofer who balances trading gains against expected fines. Our fixed scripts and imposed response do not replace those economic or equilibrium analyses.

Detection has also been studied beyond synthetic rule matching. Tao et al. [19] derive spoofing conditions from a multilevel imbalance model, calibrate it to TMX Level-2 data, and propose monitoring based on distributional distance. Chen, Hsieh, and Yang [20] use Taiwan derivatives order and trade data to study potential spoofing and a price-banding intervention. Their evidence links trading restrictions to market outcomes, while also confronting identification limits. Our uncalibrated simulator can isolate implementation effects but cannot supply comparable market evidence.

Among related working papers, Do and Putniņš [13] evaluate indicators using prosecuted cases and out-of-sample tests; Fabre and Challet [10] use Level-3 data to estimate price responses and possible spoofing gains. These studies highlight two gaps in our design: external labels and an empirically grounded response model. The new cap-threshold sweep measures sensitivity to an assumed response, not its realism.

Learning changes the problem further. Byrd [2] studies an RL trader that discovers spoofing and a learned recognizer that discourages it. Wang and Wellman [3] model adaptation between manipulation and detection. Cartea, Chang, and García-Arenas [12] examine order-book manipulation by learning algorithms. Gu et al. [17] vary liquidity and spoofing strategies, while ABIDES [4] supports richer agent interaction and communication latency. These comparisons limit the novelty claimed here to evaluation of interacting controls in a deliberately small environment.

The closest agent-compliance work is ReguSim [8], which separates a trader's reasoning, attempted orders, enforcement, and monitoring evidence. Its reported trader environment uses exogenous prices and full execution of accepted orders. Borjigin and He [9] propose constrained execution with a runtime shield; their full cryptographic proof pipeline remains future work. Kurshan, Balch, and Byrd [5] discuss agent-based governance, and Byrd [11] studies trading with LLM-generated social posts. Our study includes neither model-selected trades nor a social-information channel. The preprints are retained for these specific comparisons, alongside the published journal and conference literature.

3. Market, gates, and detectors

3.1. Market mechanics

All orders execute in the simulator, with no broker or exchange connection. Offline describes this execution setting; scripted describes the programs choosing actions. An LLM could trade in an offline simulator, but none does so in these experiments.

Component	Specification
Book and matching	One evaluated account and one instrument; integer ticks, price/time priority, partial fills
Timing	One second per attempted action; background replenishment every second; decisions every 60 seconds
Background liquidity	Three price levels per side, 10 lots per level, centered around the fundamental plus any response
Exogenous path	Fundamental starts at 1,000; increments -1, 0, +1 with probabilities 1/4, 1/2, 1/4 every 30 seconds; noise quantities 0, 1, 2 with probabilities 1/2, 1/4, 1/4 every 10 seconds
Risk and fees	100-lot worst-case position commitment; fee 0.05 tick-lots per filled lot
Sessions	720 seconds in the core study; 720 and 3,600 seconds in the throttle tests; no warm-up

The fundamental and noise streams are generated before trading begins. Within a seed and session horizon, changing a gate or the response toggle cannot consume different random draws. Unfilled market quantity expires. Background quotes are replaced between successive subject actions, so splitting orders also changes how long displayed quantities remain in the book.

Immediately before each replenishment, let B and A be total remaining buy and sell quantity, including the subject. The imbalance is $I = (B - A) / \max(1, B + A)$. In the responsive condition, background quotes shift by two ticks when I exceeds a threshold t, by minus two ticks when I is below -t, and otherwise do not shift. The unresponsive condition omits this shift. The original threshold is 0.4; the sensitivity grid varies it. This step response is stipulated, not estimated from market data.

3.2. Pre-trade gates

All conditions enforce the position-commitment bound. The legacy gate additionally rejects a limit submission above twice external reference depth, more than two resting price levels on a side, and a predicted self-match. Reference depth is the median external quantity at the two touches, excluding subject orders. The aggregate gate adds a per-side cap: remaining subject limit quantity plus the full proposed quantity must not exceed m times reference depth. Originally $m = 2$. The sensitivity grid varies this aggregate multiple while retaining the legacy per-order multiple of 2.

Both gates can throttle new limit and market submissions when $C/N > 0.9$ and N is at least 20, where C and N count subject cancellations and accepted submissions in the trailing window. The original window is 1,800 seconds. Rejected attempts do not increase N; cancellations remain available. Throttle-off conditions make the minimum submission count unreachable, without changing other gate checks. The aggregate check runs at submission and does not rebalance existing orders as external depth changes. It also checks a marketable limit at its full submitted quantity, so its rejection cost is intentionally conservative.

3.3. Pattern definitions

For a canceled order i, let q_i be submitted quantity, d_i its submission-time external reference depth, and a_i its age at cancellation. Let h_i be the fraction of q_i filled before cancellation and z_i its distance from the same-side touch at submission. For a subject fill f on the opposite side, define $Q(i,f)$ to hold when i was nonmarketable at submission, submission precedes f, f precedes cancellation, a_i is at most W, and $h_i < 0.1$. Event sequence numbers determine ordering within a timestamp. The implemented rules are:

Signal	Condition for an alert
Individual pattern	Some i and f satisfy Q(i,f), $q_i \geq 3d_i$, and $z_i \leq 3$ ticks.
Layer pattern	For one f and one displayed side, qualifying Q(i,f) orders occupy at least three distinct prices. No individual-size or distance test is added.
Aggregate pattern	For one f and displayed side, at least two Q(i,f) orders have $z_i \leq 3$ and positive remaining quantity $r_i(f)$. Their sum is at least 3 times the maximum d_i in that group.
Self-match	The subject is both buyer and seller in a fill.

Remaining quantity $r_i(f)$ is measured immediately before the shared fill, after any earlier partial executions. The conduct-alert union contains these four signals. A separate cancellation-ratio diagnostic is excluded. W is shared by the individual, layer, and aggregate patterns; changing it changes all three. These are surveillance patterns, not implementations of a complete legal test or estimates of trader intent.

4. Comparisons and experimental design

Terminal wealth is cash plus inventory valued at the terminal fundamental, after fees. For gate g, parameters p, and seed s, the response-dependent difference is D(g,p,s) = W(responsive,g,p,s) - W(unresponsive,g,p,s). The two conditions use the same gate and exogenous path. Positive D indicates a benefit from this particular response toggle; it is not a legal or general welfare measure. Executed lots, rejected actions by reason, and response activations are reported separately.

The split probe alternates sides. Each cycle submits six 10-lot orders at one price, executes two lots on the opposite side, and cancels the surviving display. Legitimate controls include small and large two-sided makers, a patient maker, a passive scheduled executor, and a 10-lot slice executor. Their objectives are assigned by construction rather than independently adjudicated. The scheduled executor is called vwap_executor in code, but has no volume forecast or VWAP benchmark.

Core comparisons use 40 seeds per arm. The cadence study reuses 40 paths across quoting schedules and rescoring windows. A subsequent parameter sample varies order count, total quantity, price spread, submission gaps, lifetime overlap, and execution delay. The original 40 draws were reused for window tuning and assigned one market seed each. Those results are exploratory and do not establish held-out recall.

Sensitivity analysis	Conditions	Episodes
Cap and response	2 scripts x 6 response thresholds x (ungated + 5 aggregate caps) x 10 seeds x 2 response states	1,440
Parameter replication	12 new vectors from the existing generator x 5 market seeds x 2 thresholds x 2 gates x 2 response states	480
Throttle and horizon	2 session lengths x (ungated + 2 gates x 4 windows + 2 gates without throttle) x 10 seeds	220

Table 1. Post-review sensitivity design: 2,140 additional episodes. The cap grid uses m in {1, 2, 3, 4, 6} and t in {0.1, 0.2, 0.25, 0.3, 0.4, 0.5}; the parameter replication uses m = 2 and t in {0.2, 0.4}. Both disable the cancellation throttle to isolate the cap-response interaction. The horizon study retains t = 0.4 and m = 2. Its windows are 120, 300, 900, and 1,800 seconds.

The sensitivity settings and parameter vectors were saved before these runs, after the earlier findings and review were known. This is not preregistration. Seeds 8000-8009 pair the cap-grid conditions, 8100-8104 pair every new parameter vector, and 8200-8209 pair the throttle conditions. Shared seeds and repeated scoring do not create independent samples. We report descriptive counts and paired means

without treating this grid as a population study. The parameter replication remains within the same authored strategy family.

5. Aggregation and the price-response boundary

Gate	Original alerts	With aggregate signal	Executed lots	Wealth
off	0/40	40/40	24.00	22.05
legacy	0/40	40/40	24.00	22.05
aggregate	0/40	0/40	24.00	-25.95

Table 2. Six-by-ten split probe over seeds 3000-3039 in the responsive market. The original throttle is enabled for both gates; the scoring window is 120 seconds. Alerts are episode counts; volume and wealth are means. Wealth is measured in tick-lots.

The split probe passes the individual-size check and occupies only one price level. Neither the individual detector nor the layer detector flags it in any of these 40 paths. The aggregate detector does flag the admitted sequence. The aggregate gate reduces admitted display and eliminates its response-dependent gain at the original settings. Agreement between a size-based gate and a size-based detector, however, is not independent evidence of protection.

The original settings make this result structurally likely. With three intact background levels of D lots per side and subject display mD on only one side, $I = m/(6 + m)$. Thus $m = 2$ gives $I = 0.25$, below 0.4. Under these static assumptions, activation requires $m > 6t/(1 - t)$; at $t = 0.4$, the boundary is $m = 4$. Trades can deplete depth before replenishment, so 0.25 is not a global bound on the simulated book. A replay diagnostic of the 40 gated parameter draws nevertheless finds no response activations and a maximum absolute imbalance of 0.282. Their identical responsive and unresponsive event streams therefore verify a restricted mechanism, not general economic prevention.

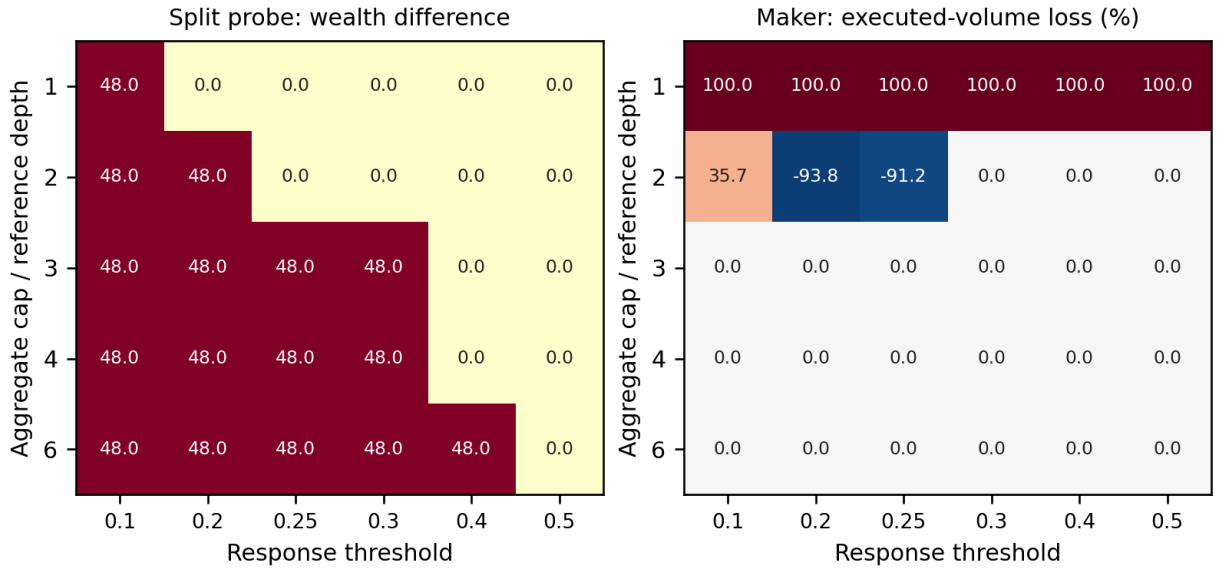


Figure 1. Cap-response sensitivity with the cancellation throttle disabled. Left: mean D for the six-by-ten split probe, in tick-lots. Right: percent loss of large-maker executed volume relative to the ungated mean at the same response threshold; negative values mean higher volume under the gate. Every cell averages the same ten seeds; colors have separate scales. Only the aggregate cap varies, while the individual limit remains twice reference depth.

At the original 20-lot aggregate cap, the split probe has mean $D = 0.00$ at $t = 0.4$. Lowering only the response threshold to 0.2 changes mean D to 48.00, positive on 10/10 paired paths. The response rule is

therefore part of the claimed protection: a cap that suppresses one assumed market response can leave gains under another. Figure 1 also shows why simply tightening the cap is not a cost-free answer. At m = 1, the 12-lot maker quotes exceed the approximately 10-lot aggregate allowance.

Threshold	Gate	Mean D	D > 0	D > 0, no alert
0.2	off	16.47	44/60	5/60
0.2	aggregate	10.96	30/60	30/60
0.4	off	11.32	27/60	0/60
0.4	aggregate	0.00	0/60	0/60

Table 3. Parameter replication: 12 vectors crossed with five shared market seeds, at cap multiple 2 with the throttle disabled. Each denominator of 60 is a set of draw-seed pairs, not 60 independent strategy families. An alert is scored at the fixed 180-second window. A positive difference is retained even if the gate rejects other actions.

At threshold 0.2, 30/60 gated draw-seed pairs retain a positive response-dependent wealth difference without a conduct alert. The same cap at threshold 0.4 leaves no positive differences in this replication. Both comparisons disable the throttle; they evaluate response sensitivity, not the full default policy. The left panel of Figure 1 is also predictable from the imposed mechanism: 48 tick-lots equals 12 cycles times two executed lots times a two-tick response when every execution benefits. The sweep makes this dependence explicit; it does not validate that response against a market.

The maker comparison is also response-dependent: at cap multiple 2 and response thresholds 0.2 and 0.25, the throttle-free gate increases executed volume. A display change alters the subsequent book response and fills, so fewer accepted orders need not mean less execution. Higher volume alone does not establish better execution quality. Crossing parameter vectors with several paths tests another source of variation, but does not validate the generator against an external strategy population. Smooth responses, varying liquidity, queue calibration, and strategic background traders remain untested.

6. Throttle timing and legitimate trading cost

The cancellation throttle counts accepted submissions. The legacy gate admits more maker quotes per decision and can reach the minimum count sooner than the aggregate gate. In the 40-seed throttle-isolation comparison, removing only the aggregate gate's throttle raises mean executed volume from 50.93 to 59.05 lots, close to the ungated 59.48. That result originally came from a 720-second session with a 1,800-second window: a triggered lockout cannot recover through expiry within the session.

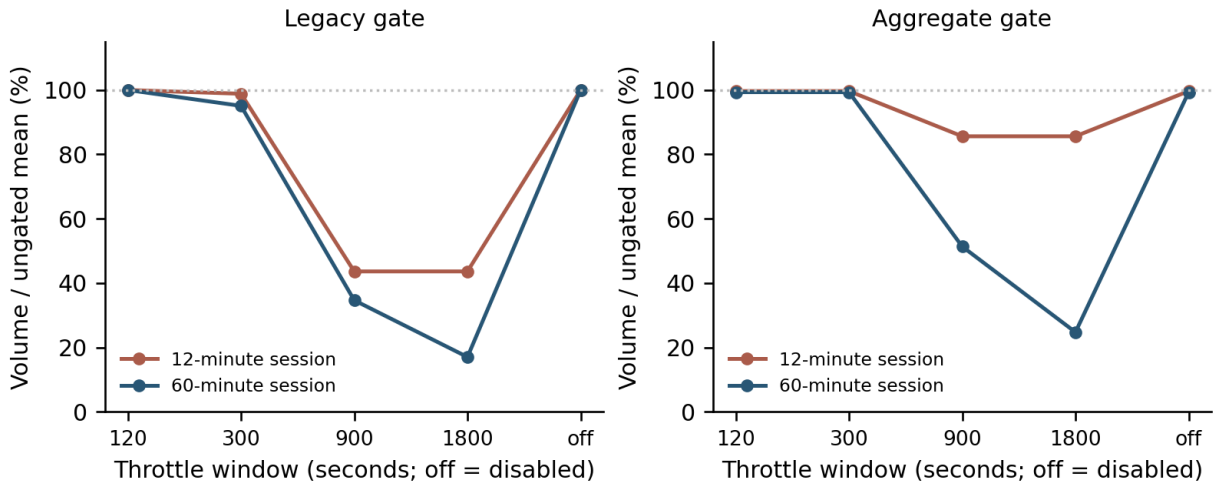


Figure 2. Mean executed volume as a percentage of the ungated mean at the same session horizon, over seeds 8200-8209. The final point disables the cancellation throttle while retaining the other gate checks. Ratios are ratios of means, not means of per-seed percentages. The two panels distinguish legacy and aggregate policies.

Minutes	Gate	Lots	Throttle rejects	Throttled	Recovered
12	legacy	26.40	26.80	10/10	0/10
12	aggregate	51.80	6.80	9/10	0/10
60	legacy	51.20	197.50	10/10	10/10
60	aggregate	74.40	169.40	10/10	10/10

Table 4. The 1,800-second throttle window at two horizons. Recovered means at least one accepted submission after the first throttle rejection, not uninterrupted trading thereafter. Throttle rejection counts are means; the last two columns count episodes.

In the one-hour condition, all ten aggregate-gated episodes resume submissions after a throttle rejection, yet mean volume is only 74.40 lots against 300.60 ungated, a 75.2% loss. The corresponding short-session aggregate arm averages 51.80 against 60.50 lots. A longer session therefore does not make the default throttle harmless; it exposes repeated periods of restricted trading.

Changing the horizon makes recovery observable, while changing the trailing window can prevent the minimum accepted-submission count from being reached at all. Recovery does not imply a small total cost: repeated lockouts can still suppress volume. These comparisons identify interactions among the window, minimum count, quoting rate, and session length. They do not establish an optimal throttle for a live market.

Legitimate task	Ungated lots	Aggregate lots	Volume loss	All rejects
Scheduled executor	27.38	27.38	0.0%	0.00
10-lot slice executor	30.12	30.12	0.0%	0.00
Small maker	31.05	31.05	0.0%	0.00
Large maker	56.50	48.55	14.1%	27.95
Patient maker, 300 s	66.28	47.60	28.2%	6.03

Table 5. Full aggregate policy, including the 1,800-second throttle, compared with no policy gate in 720-second sessions. Means over seeds 6000-6039. All rejects counts every rejected subject action, including cancellation failures; it is not the throttle-only count in Table 4.

The 14.1% and 28.2% volume losses for the two larger makers are material costs. The passive scheduled and slice executors experience no gate rejections, yet fill only 34.2% and 37.7% of their 80-lot parent orders. Zero rejection is therefore not evidence of satisfactory execution. The table measures these scripts' task completion, not a real-world false-rejection rate or a complete best-execution assessment.

7. Detector timing and legitimate counterexamples

The detector window enters Q(i,f) directly. Orders canceled after W cannot satisfy the pattern, regardless of why they were submitted. Moving a maker's re-quote cadence across W therefore changes which of its orders are eligible. Eligibility alone is insufficient: the shared fill, remaining quantity, and other conditions in Section 3.3 still have to hold.

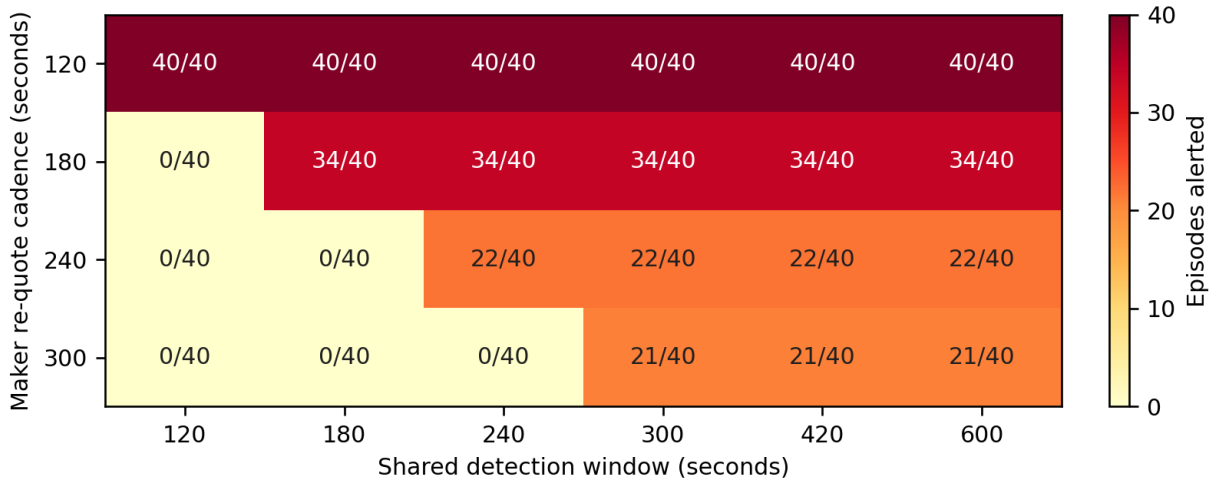


Figure 3. Rule-boundary check on an authored legitimate maker. Counts show conduct alerts among 40 ungated episodes at each cadence and shared scoring window. Each row uses the same seed cohort; each trajectory is rescored across windows. The staircase follows the rule's age condition and is not a measured operational false-positive curve.

At W = 180 seconds, a maker re-quoting every 180 seconds produces alerts in 34/40 episodes, while a 300-second cadence produces none. The maker still quotes both sides for ordinary execution. Thus the pattern can match the stated legitimate objective; it does not distinguish intent. The difference is a counterexample to evaluating specificity on a single quoting cadence, rather than a discovery that longer windows mechanically admit older orders.

The original window choice was also evaluated on the cases used to select it. Extending W from 120 to 180 seconds raised detections from 10/15 to 15/15 selected economic cases on the same parameter draws. This is tuning performance, not independent recall. Negative controls that buy only cannot meet the opposite-side-own-fill condition, and a small maker cannot challenge the aggregate size threshold. An adequate test set must exercise the rule's eligible region as well as routine behavior outside it.

Gating can remove the evidence that a detector would otherwise see. For the 180-second maker, the aggregate policy reduces mean volume from 73.80 to 58.98 lots and yields no conduct alerts. That combination should be read as smaller admitted activity with lower execution, not as an independent improvement in discrimination. Separate strategy families and independently reviewed objectives are needed to evaluate specificity beyond these examples.

8. Regulatory context

The compliance connection is to testing technical components of market-conduct supervision. For FINRA member firms, Rule 3110 [15] concerns the supervisory system and responsibilities; Notice 15-09 [14] discusses algorithm development, change control, testing, monitoring, and records. It recommends testing independently of development. Our developer-authored experiment supplies no such independent validation and does not assess a firm's supervisory organization.

FINRA Rule 5210.03 [16] includes a Type 1 disruptive-quoting pattern with multiple price levels, a supply-and-demand change, opposite-side trading, and withdrawal. The layering script resembles parts of that sequence; the same-price split probe does not reproduce the multiple-level condition. The single-account self-match check also falls short of the beneficial-ownership and related-algorithm considerations in Rule 5210.02. Notice 24-09 [6] applies existing obligations to generative-AI use; it provides context for the proposed agent study, not evidence that these scripts evaluate AI supervision.

In the U.S. derivatives context, the CFTC's guidance on CEA section 4c(a)(5)(C) makes intent to cancel before execution material and distinguishes legitimate good-faith cancellation [21]. An order pattern or an author-assigned script label cannot establish that legal intent. This framework and FINRA supervision address different regulatory settings; the simulator is not a jurisdiction-specific compliance test.

EU RTS 6 provides a further testing connection [22]. Articles 5 and 7 address development and testing methodology and separation of testing from production. Article 13 addresses surveillance adequacy, false alerts, and replay of order and transaction data. These provisions motivate examining both suspicious sequences and legitimate activity. Our logs and simulations do not demonstrate fulfillment of the regulation's organizational, surveillance, or recordkeeping obligations.

The chosen size multiples, cancellation ratios, and time windows are experimental settings. They are not prescribed by the cited regulations. Each alert requires interpretation, and an accepted sequence is not a certification of compliance.

9. Limitations and the agent study

The market has one evaluated account, one instrument, three replenishing levels, and a two-tick step response. It has no calibrated queue, exchange latency, strategic background population, auction, cross-venue routing, or beneficial-owner network. Market orders can deplete the book even when resting limits are capped. The static imbalance calculation is consequently conditional, and the cap-threshold grid is a sensitivity analysis within the same model. It does not establish real-market profitability, welfare effects, or universal prevention.

The same development process produced the strategies, labels, and detectors. Repeated market seeds test path sensitivity, not independence from that process. New parameter vectors remain variations of one authored generator. The detector window was selected after observing earlier cases, and the legitimate cadence check was motivated by that selection. The sensitivity plan was fixed after review, not before discovery. We make no population-level accuracy, power, or independent validation claim.

Even the one-hour sessions remain short and synthetic. Recovery is defined by a subsequent accepted submission, and does not imply sustained availability. Executed lots do not capture all execution quality, inventory risk, adverse selection, or opportunity cost. The tested gate includes conservative checks that can reject executable limits; observed costs belong to that implementation.

The contribution supported by these experiments is a control-evaluation case study. It supplies reproducible examples of size aggregation, response dependence, throttle interaction, and weak negative controls. Published work already treats spoofing incentives, detection, and market effects. The paper does not claim a new spoofing mechanism, a validated detector, or a generally effective defense.

A subsequent LLM study should begin only after the intended environment and control settings are fixed. It would compare model-selected attempted actions, enforcement, and monitoring against scripted baselines, while measuring legitimate task completion. Models, prompts, tool access, error handling, and analysis would need to be specified before collecting headline results. No model selected a trading action here; no customer-demand or analyst-workflow study was performed.

10. Conclusion and availability

The aggregate limit closes the illustrated order-splitting gap, but its economic effect depends on the assumed price response. Its legitimate cost also depends on how accepted-order counts interact with a cancellation throttle. Detector timing introduces a different failure: a legitimate maker can satisfy the same pattern when its cadence falls inside the scoring window. These cases support testing economic effects, recovery, and legitimate execution separately. They support a bounded methods study, with market calibration and independent strategy evaluation still required for broader claims.

10.1. Code and data availability

Code, saved results, and reproduction instructions: github.com/tokencraftdata/agentic-compliance. The repository is currently private.

10.2. Use of generative AI tools

Anthropic Claude Code assisted with the throttle-isolation, parameterized-evasion, negative-control, and detector-window-sweep modules, their tests, and results notes. OpenAI Codex assisted with the matched-counterfactual, cadence-grid, and sensitivity studies, the paper builder, literature checks, and manuscript drafting and revision. These tools were used to produce the research artifact; the reported trading experiments used fixed scripts and made no LLM calls. The human author is responsible for the methods, results, and references.

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